

Internet of Things via Kinéis Satellite Constellation

François Spies

Université of Franche-Comté – FRANCE

OUTLINE



Internet of very lonely things

KINEIS satellite constellation for the Internet of Things

Deployment plan for the KINEIS constellation in 2023

Communication with the KINEIS environment

Gpredict software for tracking satellite trajectories

Sending a message with the KIM1 board and an Arduino type microcontroller

View the signal transmitted with a software radio

The ARTIC R2 bidirectional chipset

Very Lonely Objects Goals

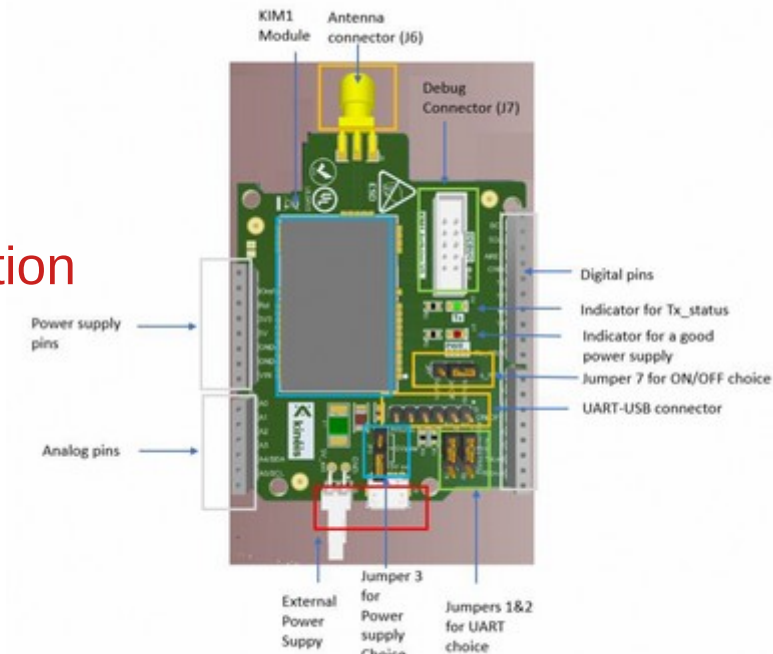
Propose an uplink and a downlink satellite communication channel

To build an object that works for a very long time

To limit the power consumption in transmission, it is necessary to have:

- A low communication rate to process weak signals (BPSK, QPSK, 200 to 400 bps)
- satellites in low orbit between 400 and 900 km (LEO Low Earth Orbit)
- period in these orbits about 1 hour 30 with passage through the poles (<https://youtu.be/XjOagUArCO0>)

To manage the object, you need a low-power microcontroller such as (ARM, STM32 ...)



Kineis Communication Module, Arduino format



Adafruit Feather ARM M0 + LoRa

Targeted Applications

Maritime

- Control of vessel movement
- Logistics monitoring, container delivery
- Boating, sea rescue

Mountain

- Hiking monitoring and mountain rescue
- Avalanche control

Agriculture, Forestry and ecological

- Long-term monitoring of activities

Volcano

- Control of volcanic activity

Wildlife and migration



Connected Containers



5 grams transmitter

Current Kineis Constellation (previously Argos)

It is designed from 8 ARGOS satellites at the end of their life (6 active including 2 RX/TX)

American meteorological satellites

- NOAA-15 (1998-810 km) and NOAA-18 (2005-860 km) : Argos-2 receiver (400bps : 7 to 32 bytes)
- NOAA-19 (2006-860 km) : Argos-3 receiver (transmit not working)

European METOP satellites

- METOP-A (2006-600 km), METOP-B (2012-850 km), METOP-C (2018- 835 km) : Argos-3 receiver (only METOP-C can transmit)

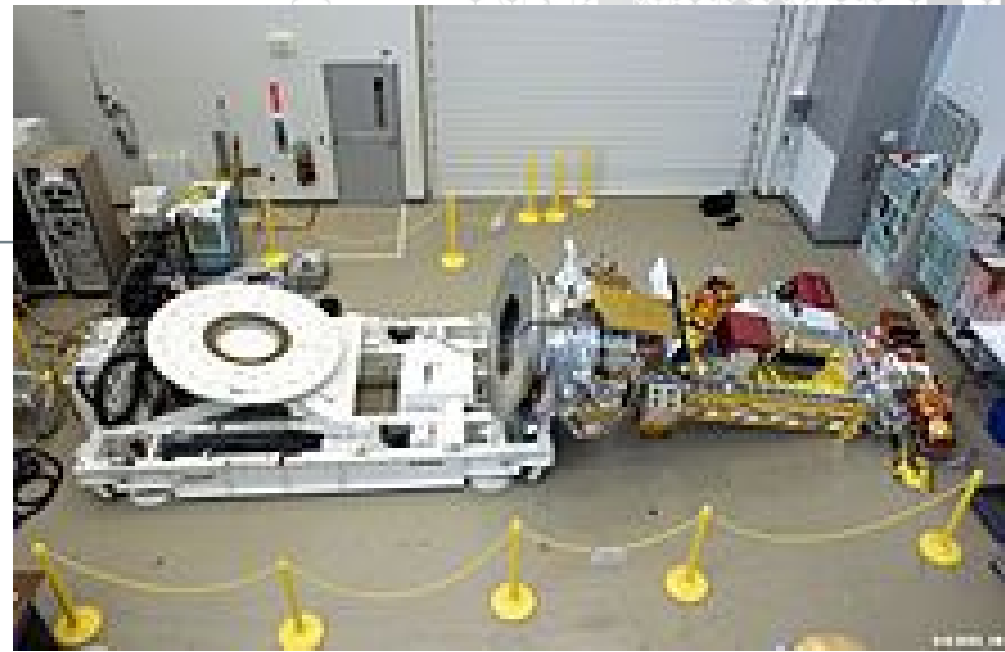
The SARAL Indian satellite

- SARAL (2013-800 km) : Argos-3 transmitter and receiver

The ANGELS test satellite

- ANGELS (2019-540 km) : Argos-4 transmitter and receiver (currently inactive)

NOAA-19 satellite



Meteorological satellite launched in February 2009

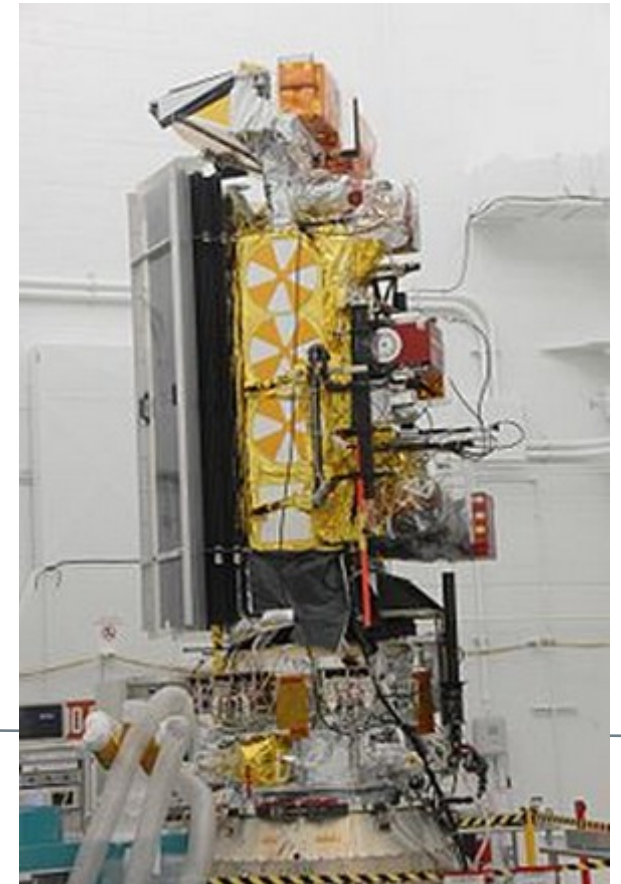
Height: 419 cm

Diameter: 188 cm

Weight: 1440 kg

Altitude: 850 km

Multiple payload: radiometer, infrared sensor, SAR, Argos



ANGELS Satellite



Launched in December 2019 to validate the concept

Dimension: 30 cm

Weight: 20 kg

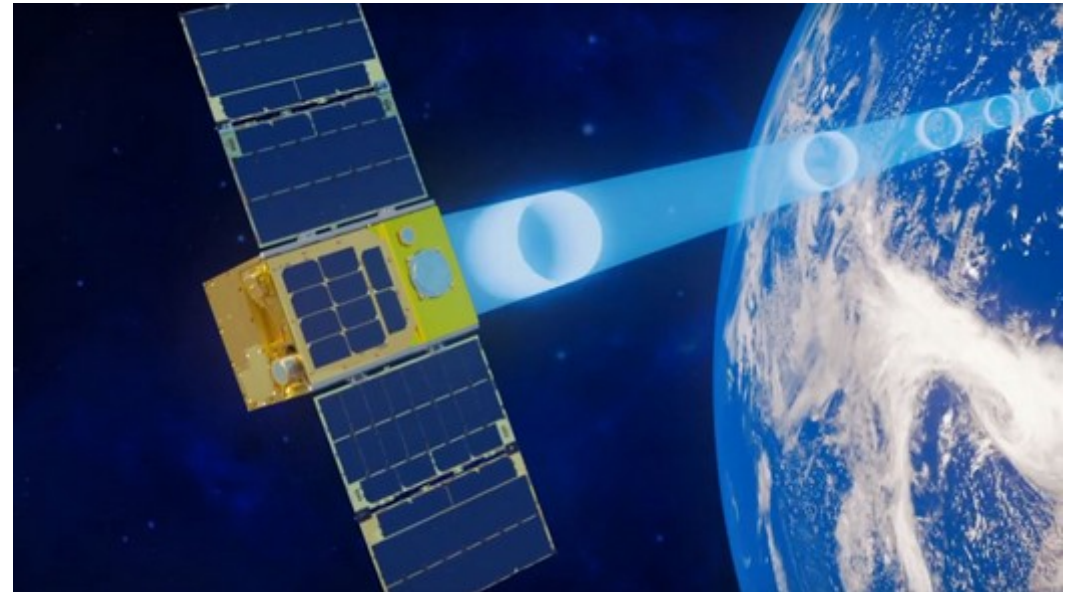
Test Duration: 2 years

Manufacture of the satellite

<https://youtu.be/CgJctFnkvoQ>

Constellation and usage

<https://youtu.be/RiMzjFHjW4k>



Future Kineis Constellation



It will be designed with 25 new satellites put into orbit in 2023

<https://youtu.be/XjOagUArCO0>

Launched by Rocket Lab, a New Zealand company

Nano-satellites will be in low orbit LEO

The average time between two passes of a point on the earth's surface will be less than 15 min

Uplink communication

- Low datarate (Argos-2/3): 401.650 MHz (+/-30 kHz) using BPSK at 400 bps
- High datarate (Argos-3) : 401.595 MHz using GMSK at 4800 bps

Downlink communication (Argos-3)

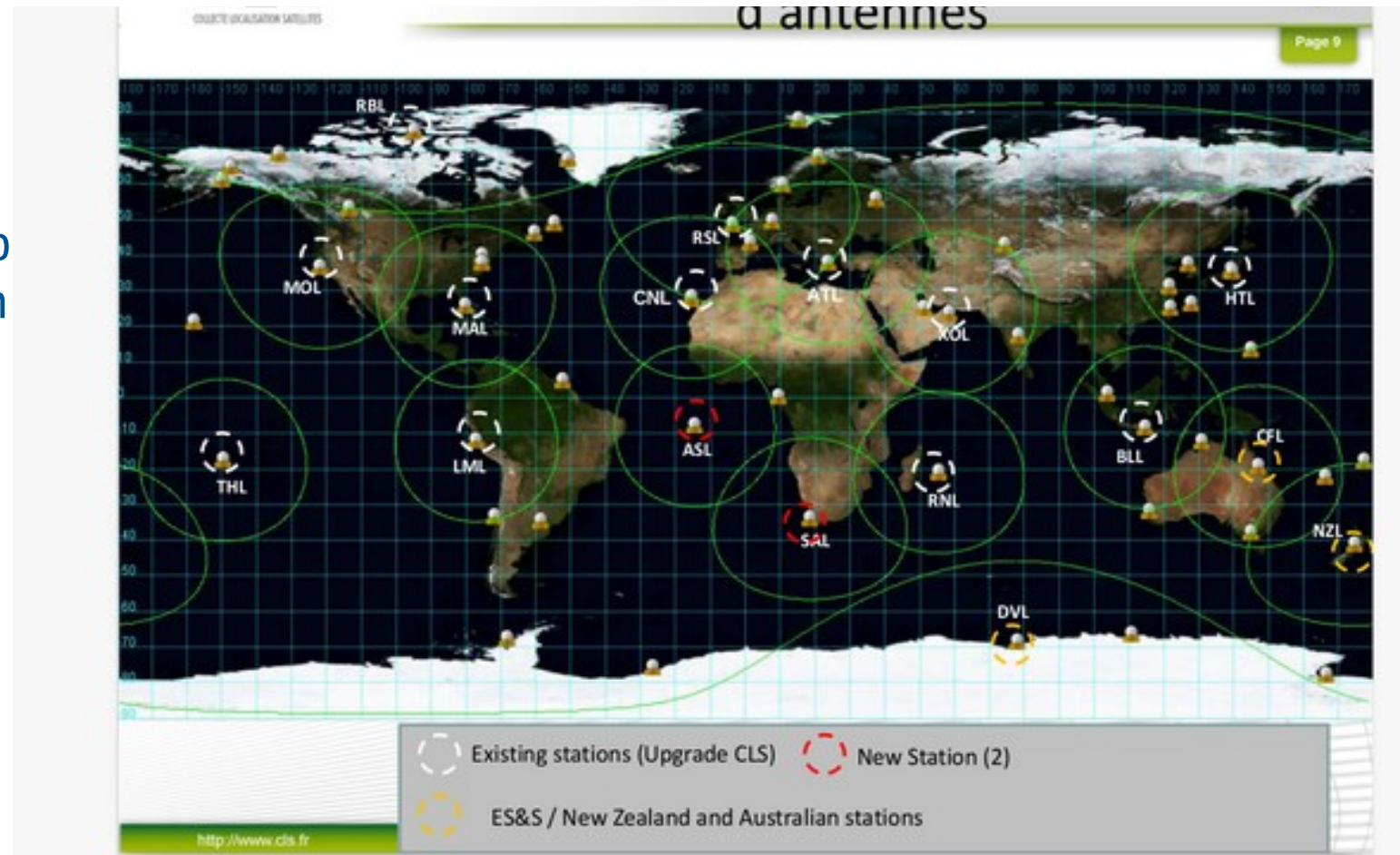
- Continuous signal at 5W (+35 dBm): 465.9875 MHz (+/-15 kHz)
- Broadcast data: time, orbit, constellation state, localization
- Dedicated data: 16 bytes and ACK

The ground Antenna segment



There are many ground stations to collect the data received by the satellites

The collected data are made available on the Web interface, by email or on an MQTT bus



Web Interface: Received data access

ARGOS

Données

Système

Support et aide

FSPIES - UTC

Messages

Critères de recherche

Sélection: Toutes

Loc et collecte

Loc seule, classe: G 3 2 1 0 A B Z

Période: Entre deux dates

Tous les messages reçus

Le plus significatif ou le meilleur

De: 28/11/2021 00:00:00

A: 28/11/2021 23:59:00

Données diagnostic:

Filtrer les positions à 1 message:

Rechercher

Exporter

N° ID	P	N° ...	D...	Date Msg ↑	Sat.	Nom du f...	Alarme	Date de loc.	Longit...	Latitude	Altitude	Ca	Vii	Qualité loc.	Frequence	Comp.	SEN...	SE...	SENSOR 03
205829		7370	440	28/11/2021 18:06:14	NP	FORMAT ...	N	28/11/2021 18:08:19	6.77969	47.47839	0.367			1	401629950.51	1	11	12	13
205829		7370	440	28/11/2021 18:07:05	NP	FORMAT ...	N	28/11/2021 18:08:19	6.77969	47.47839	0.367			1	401629950.51	1	11	12	13
205829		7370	440	28/11/2021 18:08:43	NP	FORMAT ...	N	28/11/2021 18:08:19	6.77969	47.47839	0.367			1	401629950.51	1	11	12	13
205890		7370	267	28/11/2021 18:08:56	NP	FORMAT ...	N	28/11/2021 18:08:39	6.78853	47.47985	0.393			1	401649412.05	3	BB	75	72
205890		7370	267	28/11/2021 18:11:09	NP	FORMAT ...	N	28/11/2021 18:08:39	6.78853	47.47985	0.393			1	401649412.05	1	BB	75	72
205829		7370	440	28/11/2021 18:13:34	NP	FORMAT ...	N	28/11/2021 18:08:19	6.77969	47.47839	0.367			1	401629950.51	5	11	12	13
205829		7370	393	28/11/2021 18:21:58	NK	FORMAT ...	N	28/11/2021 18:25:52	6.78602	47.47791	0.400			3	401629950.62	1	11	12	11
205890		7370	133	28/11/2021 18:24:29	NK	FORMAT ...	N				0.389				401649411.55	3	BB	75	72
205829		7370	393	28/11/2021 18:27:41	NK	FORMAT ...	N	28/11/2021 18:25:52	6.78602	47.47791	0.400			3	401629950.62	4	11	12	13
205829		7370	393	28/11/2021 18:28:31	NK	FORMAT ...	N	28/11/2021 18:25:52	6.78602	47.47791	0.400			3	401629950.62	1	11	12	13
205829		7370	195	28/11/2021 19:37:14	SR	FORMAT ...	N	28/11/2021 19:38:52	6.78186	47.47545	0.396			2	401629949.02	1	11	12	13

femto-st
SCIENCES &
TECHNOLOGIES

10/26

KIM1 : the development Kit



The Hardware part offered by Kinéis exists in several forms:

- KIM1 chipset only
- Arduino format card, STM32
- MangOH format card

This chipset can send a signal



Kineis Chipset



Arduino and STM32 board format

First use to validate the board



PC software provided by Kinéis

USB port connection
and separate power supply



Carte KIM1 format Arduino

Arduino Development

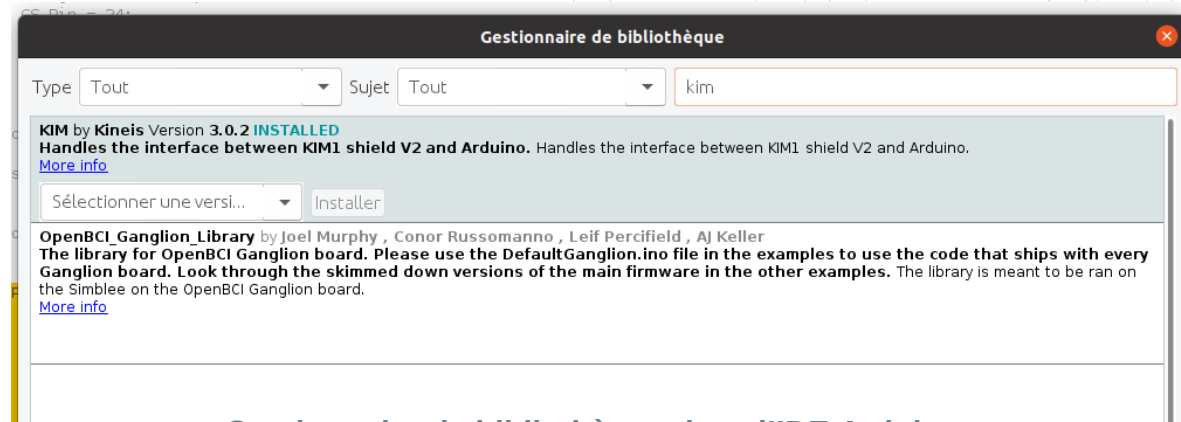
Arduino IDE

Adding the KIM library

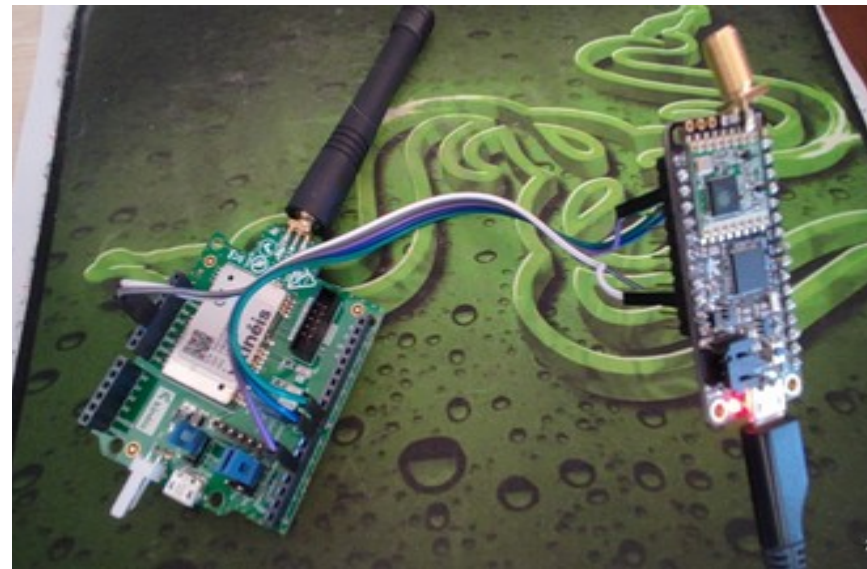
KIM1 board et jumpers configuration

Connecting KIM1 to Feather ARM

Running the KIM1 example



Gestionnaire de bibliothèque dans l'IDE Arduino



Connexion entre Adafruit Feather ARM M0/LoRa et carte satellite KIM1

Running the example

Modification of the library to Feather (pins)

using Hard Serial

LED verification

Outdoor installation

Received Data consultation on Web server



Red LED: Power good, Green LED: Transmit

```
15:24:57.225 -> Sending W
15:24:57.225 -> taille envoyee : 1
15:24:58.152 -> IRQ=C
15:24:58.152 -> duree emission (usec) : 926679
15:24:58.152 -> VBat: 4.31
15:24:58.152 -> reception active
15:25:26.995 -> No receive during
15:25:26.995 -> 28873 msec, .....
15:25:26.995 -> attente reception=28.87
15:25:26.995 -> Sending Bin to rfm95w LoRa
15:25:26.995 -> Sending W
15:25:26.995 -> taille envoyee : 1
15:25:27.922 -> IRQ=C
15:25:27.922 -> duree emission (usec) : 927246
15:25:27.922 -> VBat: 4.31
15:25:27.922 -> reception active
15:25:59.942 -> KIM -- Check fail. Please check wiring and jumpers. Freezing.
15:28:10.909 -> KIM -- Check fail. Please check wiring and jumpers. Freezing.
15:29:42.966 -> Example KIM1 Arduino shield
15:29:54.953 -> KIM -- Check fail. Please check wiring and jumpers. Freezing.
15:36:07.887 -> Example KIM1 Arduino shield
15:36:19.874 -> KIM -- Check fail. Please check wiring and jumpers. Freezing.
15:47:40.868 -> Example KIM1 Arduino shield
15:48:38.858 -> Example KIM1 Arduino shield
15:48:50.846 -> KIM -- Check fail. Please check wiring and jumpers. Freezing.
15:49:17.869 -> Example KIM1 Arduino shield
15:49:29.857 -> KIM -- Check fail. Please check wiring and jumpers. Freezing.
15:50:19.862 -> Example KIM1 Arduino shield
15:50:31.883 -> KIM -- Check fail. Please check wiring and jumpers. Freezing.
15:51:07.151 -> Example KIM1 Arduino shield
15:51:19.138 -> KIM -- Check fail. Please check wiring and jumpers. Freezing.
15:57:49.891 -> Example KIM1 Arduino shield
15:57:49.891 -> Example KIM1 Arduino shield
15:58:01.911 -> KIM -- Check fail. Please check wiring and jumpers. Freezing.
15:59:24.168 -> Example KIM1 Arduino shield
15:59:24.234 -> KIM -- Check success
15:59:24.234 ->
15:59:24.399 -> KIM -- Get ID : +ID:28,175ABAD
15:59:24.433 -> KIM -- Get SN : +SN:EDEA
15:59:24.466 -> KIM -- Get FW : +FW:KIM_HW1.3_SW1.3
15:59:24.532 -> KIM -- Get BAND : +BAND:B1
15:59:24.565 -> KIM -- Get PWR : +PWR:500
15:59:24.631 -> KIM -- Get FRQ : +FRQ:300
15:59:24.664 -> KIM -- Get TCXOWU : +TCXOWU:01388
15:59:24.731 -> KIM -- Send data ... Message sent
15:59:31.320 -> KIM -- Turn OFF
```

☒ Défilement automatique ☒ Afficher l'horodatage

Traces of the program

Satellite Passes Prediction

2 options

- Argos Web interface
- The free Software Gpredict

← → ↻ <https://argos-system.cls.fr/argos-cwi2/main.html>

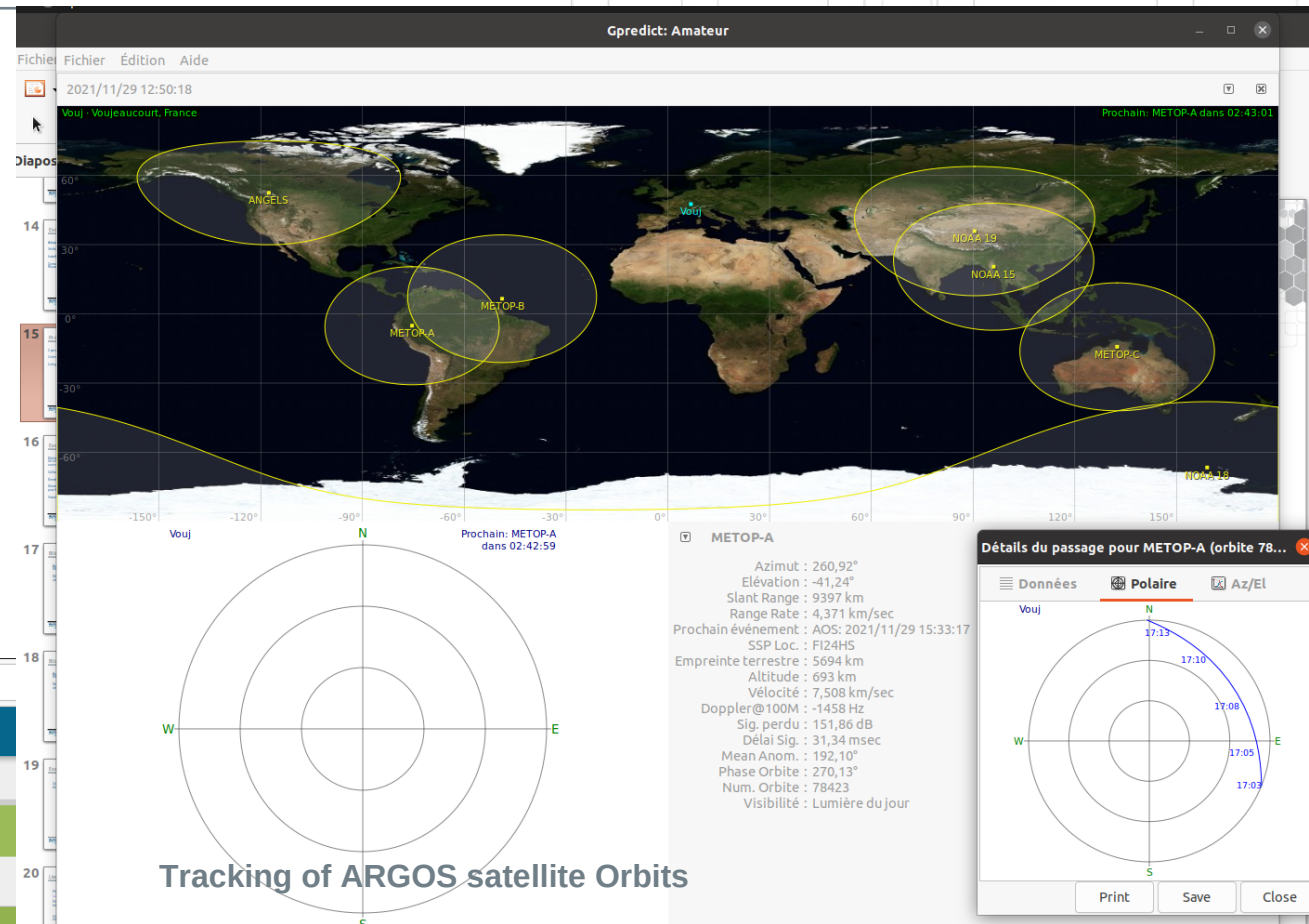
ARGOS POWERED BY CLS [Données](#) [Système](#) [Support et aide](#)

[Tableau de résultat](#) [Tableau de synthèse](#) [Affichage des blocs horaires](#)

Liste des prévisions de passage

[Exporter](#)

Satellite	Date/heure début ↑	Date/heure milieu	Date/heure fin	Durée
SR	29/11/2021 03:34:30	29/11/2021 03:39:50	29/11/2021 03:45:09	00:10:38
SR	29/11/2021 05:12:52	29/11/2021 05:18:56	29/11/2021 05:24:59	00:12:06
NP	29/11/2021 06:23:39	29/11/2021 06:29:33	29/11/2021 06:35:26	00:11:47
NK	29/11/2021 06:32:00	29/11/2021 06:37:40	29/11/2021 06:43:20	00:11:19
SR	29/11/2021 06:58:18	29/11/2021 06:59:28	29/11/2021 07:00:37	00:02:19



Practice Examples



Deploy a Kinéis + LoRa module outside and
A LoRa module inside to track execution or send data

Collect data on Kinéis MQTT broker

Transmit according to satellite passes only

Sizing a "cold" program with deep standby mode to operate
5 years in autonomy on a single battery

View the signal transmitted on a software defined radio



Outside Module Kinéis+ARM+LoRa

Feedback of Kineis Satellite Communication

Recommended transmit power: 500 mW

Transmission time around 800 ms (40 bytes at 400 bps)

One single active transmission frequency: 401.650 MHz
8 more frequencies should be added

Duty cycle requested by Kinéis: 90 seconds

At its lowest, with an elevation of 16.5 ° with METOP-C (distance greater than 2000 km)

Data transmission time from the satellite to the ground station and web server:
about 6 or 7 minutes

Argos-3 does not manage interferences in case of simultaneous receptions
Argos-4 is expected to handle simultaneous receptions

Theoretical lifetime (1/2 and 1/2)



50% of the energy for code execution and emission at 500 mW

- $250 \text{ mA} \times 0.8 \text{ sec (TX)} + 15 \text{ mA} \times 7 \text{ sec (CPU)} = 305 \text{ mA.s per transmission (85 uAh)}$

50% of the energy for deep sleep

- $305 \text{ mA.s at } 10 \text{ uA standby} = 30,500 \text{ seconds}$

For one month of operating

- $30 \times 24 \times 3,600 / 30,500 = 85 \text{ transmissions / month}$
- 1 transmission every 8.5 hours
- $85 \text{ uAh} \times 2 \times 85 = 14.5 \text{ mAh per month of operation}$

Battery sizing

- 1 year of operation: $14.5 \times 12 = 175 \text{ mAh}$
- 5 years of operation: $175 \times 5 = 875 \text{ mAh}$
- 10 years of operation: $875 \times 2 = 1750 \text{ mAh}$

On the condition of having 100% success in reception on the satellite

Theoretical lifetime (2/3 and 1/3)



66% of the energy for code execution and emission at 500 mW

- $250\text{mA} \times 0.8 \text{ sec (TX)} + 15\text{mA} \times 7 \text{ sec (CPU)} = 305 \text{ mA.s per transmission (85 uAh)}$

33% of the energy for deep sleep

- $152 \text{ mA.s at } 10 \text{ uA standby} = 15,200 \text{ seconds}$

For one month of operating

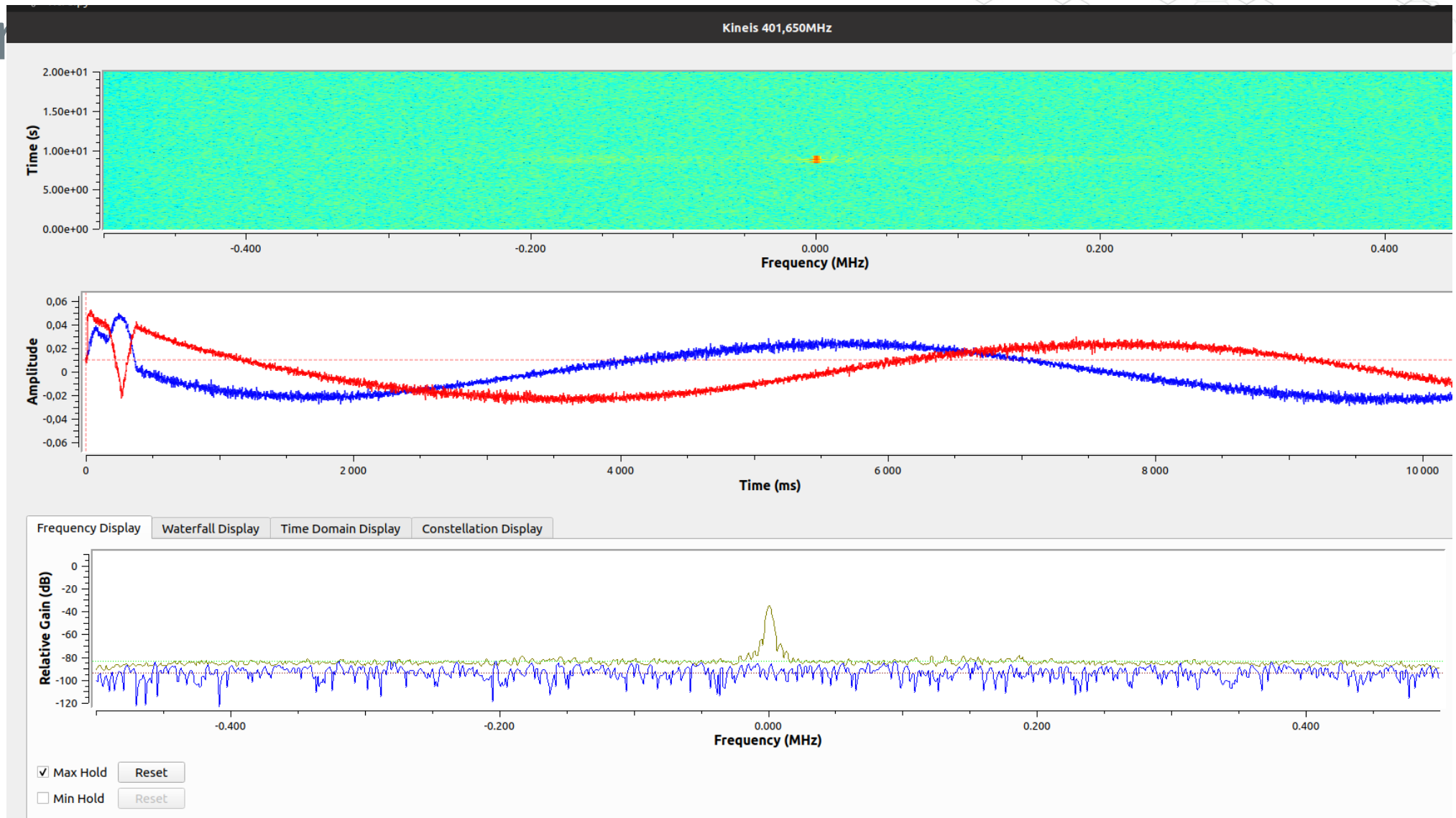
- $30 \times 24 \times 3,600 / 15,200 = 170 \text{ transmissions / month}$
- 1 transmission every 4.25 hours
- $85 \text{ uAh} \times 1.5 \times 170 = 21.7 \text{ mAh per month of operation}$

Battery sizing

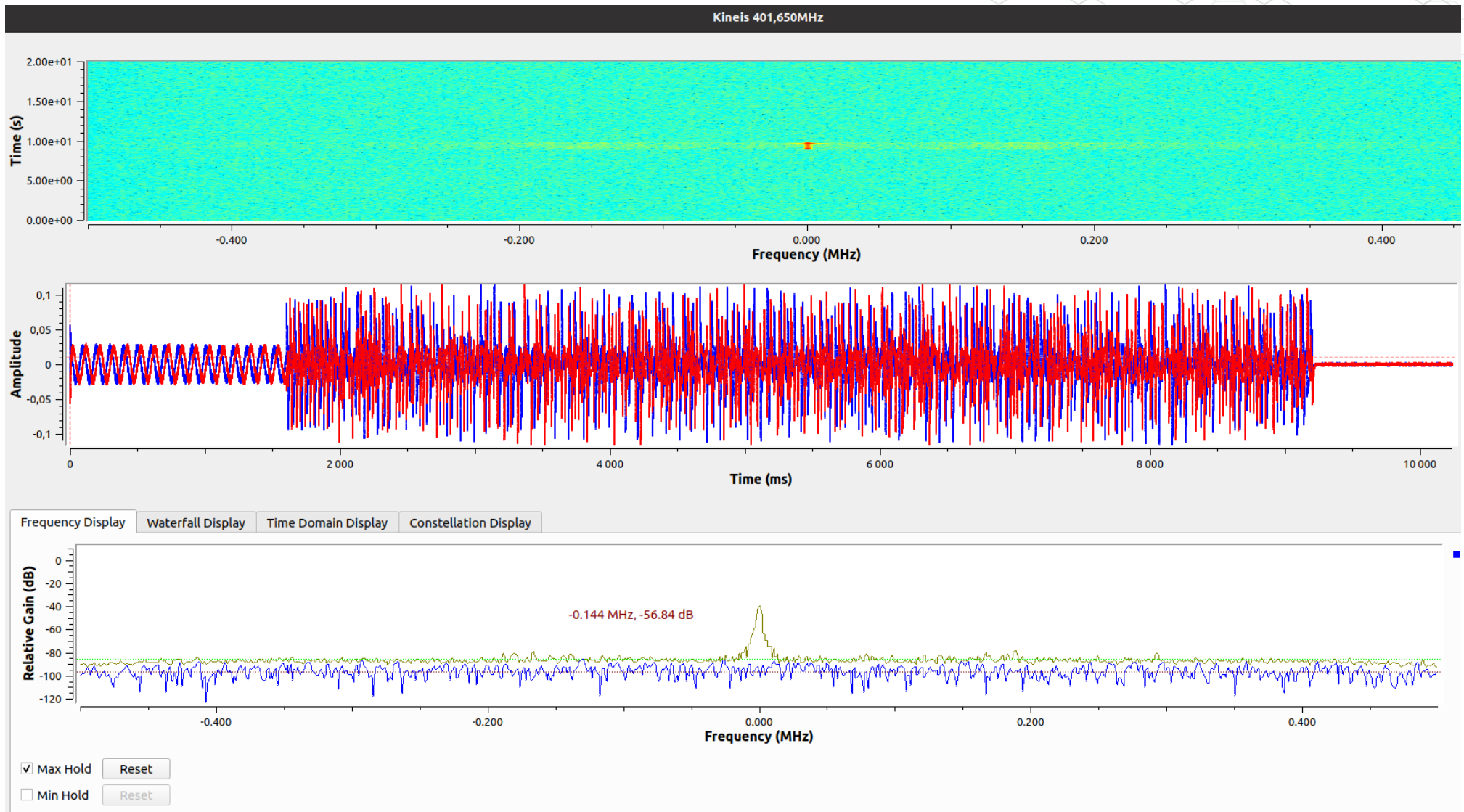
- 1 year of operation: $21.7 \times 12 = 260 \text{ mAh}$
- 5 years of operation: $260 \times 5 = 1,300 \text{ mAh}$
- 10 years of operation: $1,300 \times 2 = 2,600 \text{ mAh}$

Gnuradio/software radio: Start of the frame

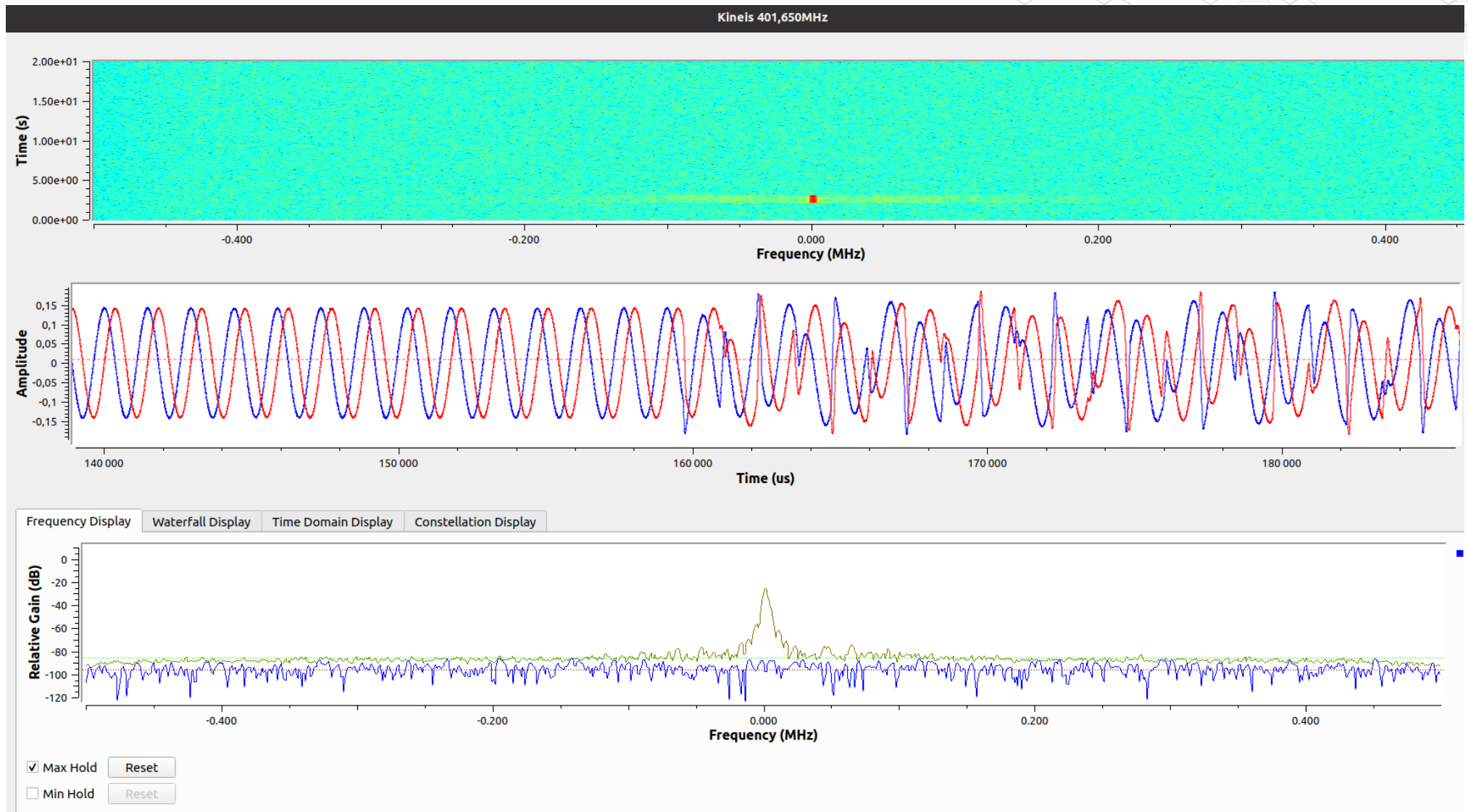
tr



Gnuradio/software radio: Argos-2 modulation



Gnuradio/SDR : from preamble to data



The ARTIC R2 new chipset

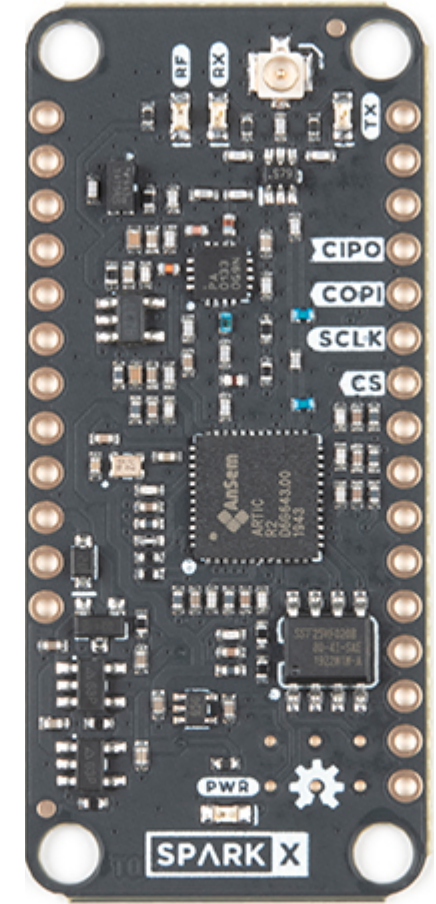
This chipset is used to send and receive signals to satellites

It accepts the following transmit standards:

- Argos-2, BPSK
- Argos-3, QPSK
- Argos-4, BPSK (Very Low Datarate, low power)
- Argos-3 and Argos-4, GMSK (High Datarate)

and receive standards:

- Argos-3, BPSK
- Argos-4, DSSS OQPSK



Sparkfun ARTIC R2

Arduino Development

Microcontroller installation: Artemis Thing plus

Library Installation: Sparkfun ARGOS ARTIC R2

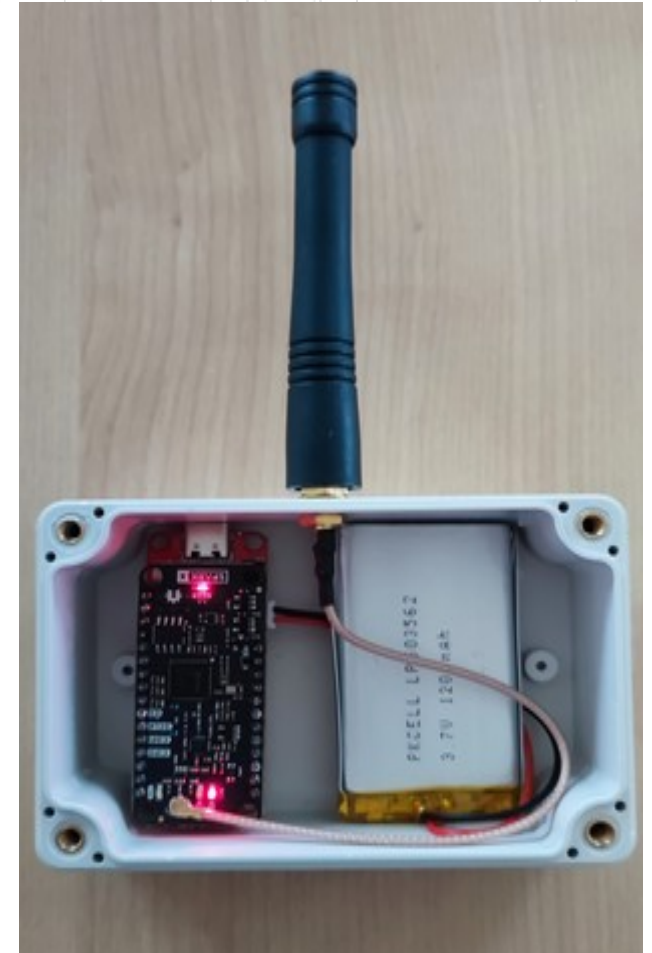
Test the examples one by one

- Firmware status
- Satellite detection
- Receive One Message
- Transmit Argos2
- Transmit Argos3

Writing of a program in progress

- Waiting for a satellite to be detected
- Sending a message (1 msg/10min or sat detected)
- Receive the acknowledgement

To transmit, you must have a known ID



Sparkfun Artemis Thing plus + Artic R2

Feedback of the ARTIC R2 *transceiver*

Very good detection of the satellite signal, including indoor (465.9875 MHz at 5 W)

Good sensitivity to receive and decoding of satellite signals

Currently, 2 satellites transmit out of 8

Transmit tests with Argos-2 signal only, satellite reception success rate is higher than KIM with lower elevations

Power consumption level seems lower than KIM
(sat. Detect or RX = 24 mA, TX = 139 mA)

CONCLUSION



Satellite constellation Internet of Things Very Lonely is a specific market with strong growth

Modern communication modes largely using on-board SDRs

Phase advance on the 2023 constellation, which makes it possible to build next generation programs which will be soon operational "full time"

Good economic model if there are a large number of objects. We can therefore hope that the costs will drop strongly in the next 3 or 4 years.
(today card 200 euros + subscription 150 euros / year)

Good potential of this specific job market for students interested in the technology

Video Links



General Presentation of Argos

https://youtu.be/sP3oQxuz_gQ

General Presentation of applications by Kinéis

<https://youtu.be/qn02Oy4N6rQ>

Visualization of the future Kineis trajectories

<https://youtu.be/XjOagUArCO0>

Container ship

